

SOFTWARE ENGINEERING

HANDS-ON SESSION

LAB-03

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SECTION: D

CENTRALIZED AUDIT TRAIL COMPLIANCE ENGINE

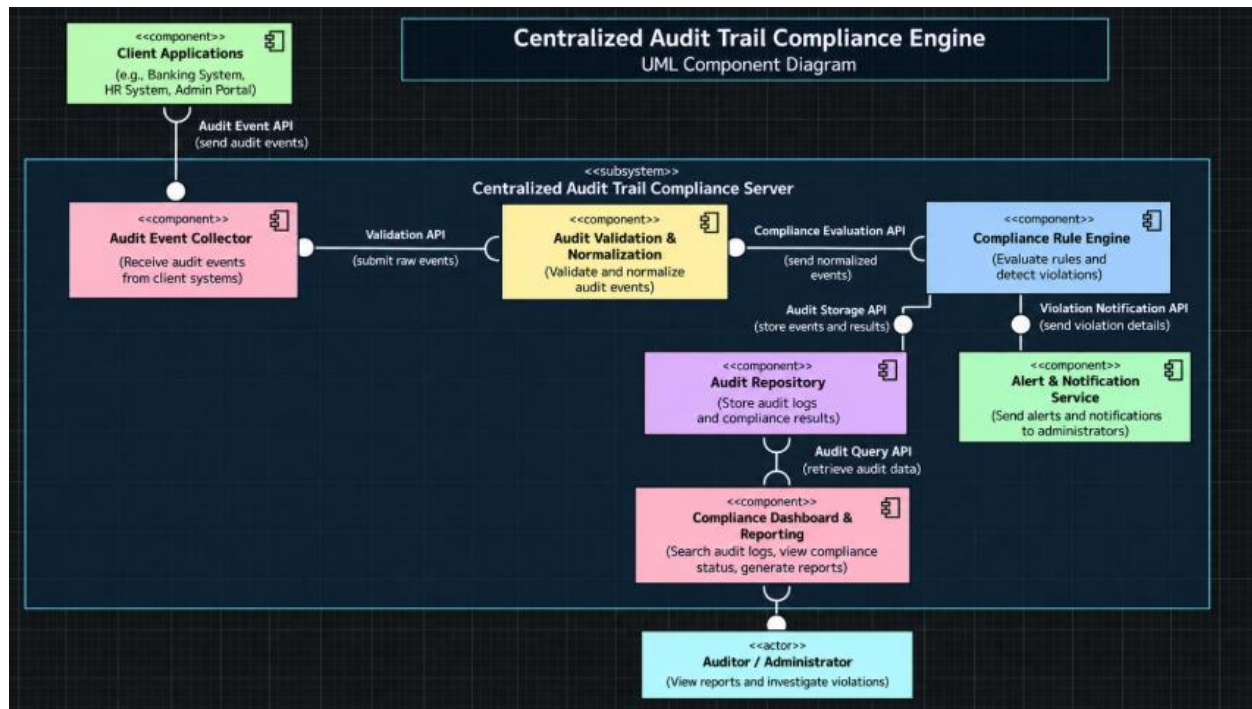
DELIVERABLE 1: UML CASE DIAGRAM

Component Diagram (PNG or PDF)

Your diagram must include at least 5 components (Adding attributes is optional).

Interface Requirements:

- Show at least 4 interfaces between components:
- Use proper UML notation for provided/required interfaces
- Show data flow and component interactions



JUSTIFICATION:

Architecture Selection:

We chose Client-Server Architecture for the Centralized Audit Trail Compliance Engine.

Architectural Choice and Reasons:

Client-Server Architecture is suitable for this system because it provides a centralized server that can receive and manage audit events generated by multiple client applications. First, it provides centralized control, allowing audit events from different applications to be collected and processed through a single centralized audit trail system. This makes it easier to manage and maintain audit information consistently. Second, it provides consistent compliance processing, because all collected audit events can be evaluated by the same Compliance Rule Engine using centrally managed compliance rules.

Security Advantage:

From a security perspective, centralizing the audit trail allows authentication and authorization to be enforced on the server. Access to sensitive audit records, compliance results, and reports can therefore be restricted to authorized users such as auditors and administrators. Centralized control also makes it easier to apply consistent security policies to audit information.

Performance Benefit:

From a performance perspective, audit collection, validation, compliance evaluation, and storage are handled by centralized server-side components. This avoids requiring every client application to independently implement and execute the complete audit-processing logic, reducing duplicated processing, and allowing the centralized components to be optimized for audit workloads.